

TREES, CITIES AND THE LANDSCAPES AROUND US

Technical Methodology, Science, Quality Control, Storytelling
Architecture & Reproducible ArcPy

LSAP Integral Ecology Atlas | Biodiversity and Conservation

Project	LSAP Integral Ecology Atlas
Map	Trees, Cities and the Landscapes Around Us
Analytical scales	Dioceses; Episcopal Conferences; LSAP members at 20 km and 100 km
Environmental foundation	Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons
Report date	3 October 2026
Status	Final analytical / publication methodology

This report documents the completed analytical workflow. Symbology and final Experience Builder presentation are intentionally outside scope.

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1. Executive summary

Trees, Cities and the Landscapes Around Us is a four-scale analytical map in the LSAP Integral Ecology Atlas. It examines how tree-dominated land cover, built-up settlement, modeled tree canopy height, population distribution, and modeled biodiversity intactness intersect with Catholic ecclesiastical geography and Laudato Si' Action Platform participation.

The map was deliberately designed as more than a tree-cover layer. Its central question is: how are Catholic communities, institutions, and LSAP participants situated within tree–settlement landscapes, and how does that relationship change as the analysis moves from an immediate local neighborhood to a broader regional context?

The analytical framework separates variables that are often conflated. Tree land-cover share measures how much of a geography is classified as trees. Modeled mean tree canopy height describes the vertical structure of tree cover and is not a canopy-cover percentage. Built-up share represents the settlement component of land cover. WorldPop provides modeled residential population context. The Biodiversity Intactness Index provides a separate modeled estimate of ecological-community intactness relative to near-undisturbed conditions.

The map also embeds the Church and LSAP into the science. Diocesan stories identify Catholics, parishes, and episcopal leadership; Episcopal Conference stories include Catholic scale, Conference leadership, LSAP enrollment, activity, goals, and participating sectors; member stories begin with the participating institution itself, then explain its local and regional landscape, science, scale relationship, and stewardship possibilities.

2. Final publication layers

Feature class	Map alias	Scale	Features	Final user fields
TreesCitiesLandscapes_Dioceses_V2	Dioceses — Tree–Settlement Landscapes	Diocese	2,874	59
TreesCitiesLandscapes_C	Episcopal Conferences —	Episcopal Conference	122	74

onferences_V2	Tree–Settlement Landscapes			
TreesCitiesLandscapes_Members_20km_V2	LSAP Members — 20 km Local Ecological Neighborhood	LSAP member / 20 km	2,739	61
TreesCitiesLandscapes_Members_100km_V2	LSAP Members — 100 km Regional Landscape Context	LSAP member / 100 km	2,739	61

2.1 Map group

The final layers are organized in the ArcGIS Pro map group “Trees, Cities and the Landscapes Around Us” in this order:

- LSAP Members — 20 km Local Ecological Neighborhood
- LSAP Members — 100 km Regional Landscape Context
- Dioceses — Tree–Settlement Landscapes
- Episcopal Conferences — Tree–Settlement Landscapes

2.2 Publication purpose by scale

Scale	Primary role
Diocese	Place-based Church geography; science + Catholics/parishes + bishop/sede vacante + locally relevant stewardship possibilities.
Episcopal Conference	Broad tree–settlement pattern; Catholics/parishes + Conference president + LSAP activity; broad enabling responses and direction to closer scales.
LSAP 20 km	Immediate ecological neighborhood around a participating institution/community; locally relevant stewardship context.
LSAP 100 km	Wider regional landscape; explains how the member’s ecological setting changes beyond its immediate neighborhood.

3. Research question and design principles

The map is built around five linked questions:

22. Who is here? Which Catholic jurisdiction or LSAP participant is being described?
23. What kind of tree–settlement landscape surrounds that Church community or institution?
24. What do the environmental indicators say about tree structure, people, and biodiversity?
25. How does the picture change with scale, especially for LSAP members between 20 km and 100 km?
26. What stewardship directions are scientifically reasonable at this scale without over-prescribing local action?

Key design principles:

- Identity before statistics: stories begin with the diocese, Conference, or LSAP member.
- Science before solutions: stewardship language follows measured/modelled context.
- No ecological moralizing: low tree cover is not automatically a deficit; naturally open ecosystems are explicitly protected from tree-planting assumptions.
- Scale matters: Episcopal Conference guidance is broad and enabling, while diocesan/member narratives can discuss locally relevant stewardship possibilities.
- Data coverage is explicit: incomplete classification changes how percentages are interpreted and how stories are worded.
- Raw source fields remain untouched where possible; publication-ready derived fields carry explicit units.

4. Source data and scientific provenance

4.1 Catholic ecclesiastical geography

Diocesan and Episcopal Conference geometries come from the project's versioned Catholic jurisdictional framework. The analytical source feature classes used here were LSAP_Diocese_BiodiversitySummary and LSAP_EpiscopalConferences_BiodiversitySummary.

Diocesan attributes include jurisdiction type/name, Catholics, parishes, clergy statistics, bishop/ordinary information, and sede vacante context. Conference attributes include Catholic/parish aggregates, president information, and LSAP aggregate activity fields.

4.2 Laudato Si' Action Platform member data

The member analyses use LSAP_MemberPoints_20km_BiodiversitySummary and LSAP_MemberPoints_100km_BiodiversitySummary. Both contain 2,739 publishable/geocodable member records. BUFFER_ID is the authoritative pairing key because it is unique and identical across both scales; Account_Name is not used as a join key because duplicate member names occur.

4.3 Esri Global Biodiversity and Conservation Hexagons

The environmental source is Esri's Global Biodiversity and Conservation Hexagons in ArcGIS Living Atlas. Esri's 2026 update describes a nested H3 product with 82 harmonized summary statistics, all linear/aerial units standardized in meters, and improved weighted treatment of partial pixels for mean and predominant statistics. Land cover is based on Esri Sentinel-2 10 m Land Use/Land Cover.

The 2025 introduction identifies H3 Levels 2–5, ranging from approximately 87,000 km² to 250 km² per hexagon, and lists Tree Canopy Height, Biodiversity Intactness, WorldPop, IUCN Red List, land cover, protected areas, carbon, terrain, and other environmental sources among the summarized themes.

Source: [Esri — Global Biodiversity and Conservation Hexagons... but better! \(2026\)](#) | [Esri — Introducing Global Biodiversity and Conservation Hexagons \(2025\)](#)

4.4 Tree canopy height: WRI / Meta

The tree-structure indicator in the hexagons is sourced from the High Resolution Canopy Height Map distributed by World Resources Institute's Land & Carbon Lab. WRI describes the product as global sub-meter canopy-height maps produced by a machine-learning model trained on aerial LiDAR and RGB imagery and applied to very-high-resolution satellite imagery.

Important source cautions retained in this project: imagery dates and atmospheric conditions vary; the model was trained/validated over a limited geographic range; field validation for very tall trees above 25 m was not performed; and terrain slope may introduce systematic bias through shadow geometry. Therefore the derived CANOPY_MEAN is treated as modeled structural context, not a direct measurement of every tree.

Source: [WRI Data Explorer — High Resolution Canopy Height Map](#)

4.5 Biodiversity Intactness Index: Natural History Museum / PREDICTS

The Biodiversity Intactness Index (BII) is a modeled indicator of how much of a region's original ecological community remains on average despite human pressures. The Natural History Museum describes the index as combining modeled total abundance with compositional similarity to near-undisturbed sites, using the PREDICTS evidence base.

BII is not a simple 'ecosystem health percentage' and should not be interpreted as a literal count of species remaining. The Natural History Museum notes that values above 90% fall within the proposed planetary-boundary range.

Source: [Natural History Museum — Biodiversity Intactness Index](#)

4.6 WorldPop

WorldPop provides modeled gridded residential population estimates. WorldPop’s Global2 methods disaggregate census/projection-linked population counts to high-resolution grid cells using geospatial covariates and machine-learning methods. The project uses TOTAL_WorldPop_2025 summarized through the environmental hexagons, then derives a represented population density from the analysis geography.

WorldPop values are not substituted for Catholic population counts. The report and story text explicitly preserve the fact that WorldPop and Church statistics are different datasets produced for different purposes.

Source: [WorldPop — Gridded population estimate datasets and tools](#) | [WorldPop Global2](#)

5. Environmental hexagon summarization and weighted statistics

The four analytical source layers were previously created by relating Catholic or member geographies to Esri’s Level-5 H3 environmental summaries. For continuous or percentage indicators, partial overlaps must not be treated as equal observations; a small hexagon fragment should not carry the same influence as a nearly complete overlap.

Area-weighted mean	$\bar{x}_w = \Sigma(x_i \times A_i) / \Sigma(A_i)$ x_i is the source H3 value and A_i is the hexagon fragment area inside the Church/member geography. Null values are excluded from both numerator and denominator.
Overlap fraction	$f_i = A_{overlap,i} / A_{hex,i}$ Used when a hexagon-wide amount is prorated to the intersected portion, only when that assumption is scientifically defensible.

The 2026 Esri update is methodologically aligned with this concern: Esri reports improved weighted handling of partial pixels within H3 cells, which improves mean and predominant statistics.

This report distinguishes first-stage source summarization (original environmental datasets → Esri H3 hexagons) from second-stage project summarization (H3 hexagons → dioceses, Conferences, and member analysis geographies).

6. Core variables and derived fields

Field	Meaning	Analytical role
PCT_LULC_Trees / TREE_LAND_PCT	Tree-classified land-cover share of the represented geography	Primary tree amount
PCT_LULC_Built / BUILT_PCT	Built-up land share	Primary settlement amount
PCT_LULC_Veg / OTHER_VEG_PCT	Other vegetation share	Supporting vegetation context
TREE_VEG_PCT	Trees + Other Vegetation	Descriptive combined vegetated share; not used as the classification backbone
WAVG_Tree_Canopy_Mean / CANOPY_MEAN	Modeled mean canopy height	Supporting tree-structure science
TOTAL_WorldPop_2025 / POP_2025	Modeled 2025 population	Human settlement context
POP_DENSITY	WorldPop / represented km²	Density context
WAVG_Biodiversity_Intactness_M / BIODIV_INTACT	Modeled BII	Supporting ecological-community context
VALID_COVER_PCT	Sum of eight meaningful land-cover classes	Coverage denominator
TREE_CLASS_PCT	Tree share normalized to classified cover	Tree-settlement classification and cross-scale comparison

BUILT_CLASS_PCT	Built share normalized to classified cover	Tree–settlement classification and cross-scale comparison
Represented population density	$\text{POP_DENSITY} = \text{POP_2025} / \text{LANDSCAPE_KM2}$ People per represented analysis km ² ; contextual only, not interchangeable with Catholic population.	
Tree + other vegetation	$\text{TREE_VEG_PCT} = \text{TREE_LAND_PCT} + \text{OTHER_VEG_PCT}$ Descriptive vegetated context; the map title and primary classification remain focused on trees and settlement.	

7. Area-unit safeguards and coverage model

Area handling is a critical methodological control. The four source layers store ANALYSIS_AREA or BUFFER_AREA and TOTAL_LULC_* values at square-meter scale. Earlier atlas analyses demonstrated the risk of accidentally labeling raw m² values as km² or converting an already-converted field twice.

This workflow does not overwrite raw source area fields. Instead it diagnoses source magnitude and writes new explicit km² fields. LANDSCAPE_KM2, TREE_AREA_KM2, BUILT_AREA_KM2, VEG_AREA_KM2, and TREE_VEG_AREA_KM2 are publication-ready derivatives.

Area conversion	$\text{Area_km}^2 = \text{Area_m}^2 / 1,000,000$ Applied exactly once after a median-magnitude unit diagnostic.
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7.1 Coverage

Cloud and NoData are coverage conditions, not ecological land-cover classes. Eight valid classes are used to define classified coverage: Water, Trees, Cropland, Built, Rangeland, Bare, Snow/Ice, and Other Vegetation.

Valid classified coverage	$\text{VALID_COVER_PCT} = \sum \text{eight valid land-cover percentages}$ The fraction of the represented geography described by meaningful land-cover classes.
Unclassified coverage	$\text{UNCLASSIFIED_PCT} = \max(0, 100 - \text{VALID_COVER_PCT})$ Total area not represented by the eight ecological/land-use classes.
Cloud + NoData	$\text{CLOUD_NODATA_PCT} = \text{PCT_LULC_Cloud} + \text{PCT_LULC_NoData}$ Separates known source-classification gaps.
Other unmapped residual	$\text{OTHER_UNMAPPED_PCT} = \max(0, 100 - (\text{VALID_COVER_PCT} + \text{CLOUD_NODATA_PCT}))$ Needed especially at 100 km member scale, where some records contained additional unrepresented source coverage.

8. Tree–settlement landscape classification

Tree–settlement classes are project-derived descriptive classes. They are not official Esri, WRI, or biodiversity-status categories. Classification uses tree and built shares normalized to the classified portion of the geography so incomplete source coverage does not distort the category.

Normalized tree share	$\text{TREE_CLASS_PCT} = 100 \times \text{TREE_LAND_PCT} /$
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	VALID_COVER_PCT Used for classification and 20→100 km comparison.
Normalized built share	BUILT_CLASS_PCT = $100 \times \text{BUILT_PCT} / \text{VALID_COVER_PCT}$ Used for classification and 20→100 km comparison.

8.1 Component thresholds

Dimension	Class	Threshold
Tree	Tree-Rich	TREE_CLASS_PCT ≥40%
Tree	Mixed Tree Cover	$15\% \leq \text{TREE_CLASS_PCT} < 40\%$
Tree	Lower Tree Cover	TREE_CLASS_PCT <15%
Settlement	Lightly Built	BUILT_CLASS_PCT <5%
Settlement	Settled	$5\% \leq \text{BUILT_CLASS_PCT} < 15\%$
Settlement	Highly Built	BUILT_CLASS_PCT ≥15%

8.2 Final combined classes

Final class	Rule	Interpretive intent
Tree-Dominated Landscape	Tree-Rich + Lightly Built	Trees dominate a relatively lightly developed geography.
Tree-Rich Settled Landscape	Tree-Rich + Settled/Highly Built	Substantial tree cover coexists with settlement.
Mixed Tree–Settlement Landscape	Mixed Tree Cover + Lightly Built/Settled	Tree, settlement, and other land systems coexist.
Built Landscape with Substantial Tree Cover	Mixed Tree Cover + Highly Built	Development is prominent but trees remain substantial.
Built Landscape with Limited Tree Cover	Lower Tree Cover + Highly Built	Built-up land is prominent and tree share is limited.
Open or Sparsely Treed Landscape	Lower Tree Cover + Lightly Built/Settled	Tree cover is limited; native open ecosystems or other land systems may dominate.

Low tree cover is never interpreted automatically as degradation. In grasslands, shrublands, wetlands, drylands, alpine systems, or some agricultural settings, more trees may be ecologically inappropriate.

9. Modeled tree canopy height interpretation

CANOPY_MEAN is the summarized modeled mean tree canopy height, not percent canopy cover. The project adds descriptive text classes to make the model output understandable in popups. These are project interpretation classes, not WRI-defined categories.

Project class	Modeled mean height
Short Canopy Structure	<5 m
Lower Canopy Structure	5–<10 m
Moderate Canopy Structure	10–<15 m
Tall Canopy Structure	15–<20 m
Very Tall Canopy Structure	20–<25 m
Exceptionally Tall Modeled Canopy	≥25 m

Values ≥ 25 m receive special caution because WRI states that field validation for very tall trees above 25 m was not performed during development of the global dataset. The project also retains WRI's terrain-slope caution.

Example interpretation: 24.5 m is described as a very tall modeled canopy structure, not as a direct measurement that every tree is 24.5 m tall.

10. Biodiversity Intactness Index interpretation

BIODIV_INTACT is a modeled mean Biodiversity Intactness Index from the environmental source. The Natural History Museum describes BII as combining modeled abundance and community similarity relative to near-undisturbed reference conditions.

Project class	BII range	Interpretation
High Modeled Biodiversity Intactness	≥ 0.90	High modeled intactness; ≥ 0.90 is also associated with the proposed planetary-boundary range in NHM material.
Relatively High Modeled Biodiversity Intactness	$0.80 - < 0.90$	Much of the modeled original ecological-community signal remains, with meaningful human-pressure effects.
Intermediate Modeled Biodiversity Intactness	$0.70 - < 0.80$	Substantial modification relative to near-undisturbed reference conditions.
Reduced Modeled Biodiversity Intactness	$0.60 - < 0.70$	Stronger modeled departure from near-undisturbed reference conditions.
Substantially Reduced Modeled Biodiversity Intactness	< 0.60	Large modeled departure from near-undisturbed reference conditions.

A value such as 0.86 is not narrated as “86% healthy.” Instead, the story explains that it represents roughly 86% of the modeled original ecological-community signal remaining on average, while still reflecting meaningful ecological change from near-undisturbed reference conditions.

Values slightly above 1.0 are treated as near-reference modeled values rather than “more than 100% intact.”

11. WorldPop human-settlement context

POP_DENSITY is calculated from the WorldPop population total summarized through the analysis geography and LANDSCAPE_KM2. It is used to describe the human intensity of the landscape, not to calculate Catholic participation rates.

In story text, WorldPop and Catholic statistics remain explicitly separate: e.g., a diocese may be described as containing approximately 850,000 Catholics across 59 parishes, while its environmental population context is approximately 12,415 modeled residents per represented analysis km².

12. LSAP member 20 km → 100 km transition analysis

The member analysis is paired 1:1 using BUFFER_ID. Changes are calculated from normalized classified-cover shares rather than raw full-geography percentages, preventing incomplete 100 km coverage from falsely appearing as ecological decline.

Tree change	$\text{TREE_CHANGE_PP} = \text{REGION_TREE_CLASS_PCT} - \text{LOCAL_TREE_CLASS_PCT}$ Positive values mean trees become more prominent within classified cover at the 100 km scale.
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Built change	$\text{BUILT_CHANGE_PP} = \text{REGION_BUILT_CLASS_PCT} - \text{LOCAL_BUILT_CLASS_PCT}$ Negative values mean built-up land becomes less prominent within classified cover as the spatial context expands.
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12.1 Comparison confidence

Class	Minimum classified coverage across both scales
High-Confidence Comparison	≥90%
Moderate-Confidence Comparison	≥75% and <90%
Limited-Coverage Comparison	≥50% and <75%
Very Limited-Coverage Comparison	<50%

For limited-coverage comparisons, the popup states that the direction of change is more reliable than the exact magnitude.

13. Church and LSAP storytelling architecture

The final narrative order is intentionally human-centered: identity first, then landscape, then science, then scale relationship, then stewardship, followed by a separate technical note.

13.1 Diocese

- Who: jurisdiction name/type, Catholics, parishes, bishop or sede vacante.
- Landscape: tree–settlement class, tree share, built share, represented km².
- Science: WorldPop density, modeled canopy-height class, BII class.
- Stewardship: class-specific but general enough to require local ecological assessment.
- Technical note: model/caveat information kept separate from the main integrated public story.

13.2 Episcopal Conference

- Who: Conference president, Catholics, parishes.
- LSAP: enrolled participants, plans/actions/reflections when present, top related Laudato Si' goals, and leading participating sectors.
- Landscape and science: same broad environmental interpretation as other scales.
- Solutions: broad enabling/coordination roles only; no site-level prescriptions.
- Explore at a Closer Scale: directs readers to dioceses and member ecological neighborhoods for place-specific stewardship.

13.3 LSAP members

- Who: member name, sector, country.
- Landscape: local 20 km or regional 100 km tree–settlement context.
- Science: population context, canopy-height class, BII class.
- Scale relationship: explicit 20 km → 100 km change in tree and built shares where coverage allows.
- Stewardship: local-scale possibilities at 20 km; partnership/regional-system possibilities at 100 km.

14. Scale-specific stewardship framework

Stewardship language is derived from the tree–settlement class, but it is deliberately framed as possibilities rather than priorities or prescriptions. Every solution block includes ecological caveats appropriate to the class.

Class	Typical stewardship emphasis
Tree-Dominated	Protect mature wooded systems; connectivity; watershed function; native tree diversity; responsible woodland management.
Tree-Rich Settled	Protect mature canopy; succession planning; shade; stormwater; native species; connect campuses and surrounding green networks.
Mixed Tree–Settlement	Maintain ecological connections; native vegetation; soils/water; institutional land as a demonstration landscape.
Built + Substantial Trees	Protect existing urban/institutional canopy; succession; shade; stormwater; heat mitigation; habitat links.
Built + Limited Trees	Locally appropriate greening; protect mature trees; heat/stormwater; native vegetation; equitable green space.
Open/Sparsely Treed	Understand native open ecology first; prioritize grasslands, shrublands, wetlands, drylands, soils, agriculture, or water where appropriate.

15. Quality-control record

The final workflow uses a limited set of high-value scientific and publication checks rather than repeated large QC loops. The executed outputs below are retained as the evidence base for the final analytical choices.

15.1 Scientific distribution audit

Retained QC output

STEP 1 – SCIENTIFIC DISTRIBUTION + RELATIONSHIP AUDIT

DIOCESES

Tree land-cover share:

10th 3.33 | 25th 13.96 | median 34.34 | 75th 55.96 | 90th 74.43

Mean canopy height:

10th 2.50 | 25th 6.37 | median 11.09 | 75th 16.49 | 90th 21.50

Built-up share:

10th 0.33 | 25th 1.23 | median 4.26 | 75th 10.25 | 90th 20.74

WorldPop density:

10th 9.26 | 25th 30.25 | median 85.95 | 75th 241.48 | 90th 587.69

Biodiversity intactness:

10th 0.76 | 25th 0.82 | median 0.88 | 75th 0.93 | 90th 0.96

Coverage:

<95% = 57 | <90% = 22

Relationships:

Trees vs canopy $r=0.852$

Trees vs built $r=-0.206$

Canopy vs built $r=0.017$

Trees vs intactness $r=0.552$

EPISCOPAL CONFERENCES

Tree share median 37.20 | Canopy median 11.38 | Built median 3.47

WorldPop density median 75.23 | BII median 0.88

Coverage <95% = 2 | <90% = 1

Relationships:

Trees vs canopy $r=0.890$

Trees vs built $r=-0.129$

Canopy vs built $r=0.030$

Trees vs intactness $r=0.516$

LSAP MEMBERS – 20 KM
 Tree share median 13.35 | Canopy median 12.09 | Built median 34.25
 WorldPop density median 1,030.86 | BII median 0.82
 Coverage <95% = 259 | <90% = 140
 Relationships:
 Trees vs canopy $r=0.638$
 Trees vs built $r=-0.452$
 Canopy vs built $r=0.114$
 Trees vs intactness $r=0.555$

LSAP MEMBERS – 100 KM
 Tree share median 25.05 | Canopy median 11.77 | Built median 9.72
 WorldPop density median 241.29 | BII median 0.86
 Coverage <95% = 1,106 | <90% = 850
 Relationships:
 Trees vs canopy $r=0.801$
 Trees vs built $r=-0.148$
 Canopy vs built $r=0.158$
 Trees vs intactness $r=0.469$

15.2 Area-unit QC

Retained QC output

STEP 2 – AREA UNIT CHECK

DIOCESES

Median stored ANALYSIS_AREA: 11,516,489,885.973
 Detected units: M2
 Publication km² fields divide by 1,000,000 exactly once.

EPISCOPAL CONFERENCES

Median stored ANALYSIS_AREA: 280,242,973,406.825
 Detected units: M2

LSAP MEMBERS – 20 KM

Median stored BUFFER_AREA: 1,256,636,981.225
 Detected units: M2

LSAP MEMBERS – 100 KM

Median stored BUFFER_AREA: 22,556,872,817.134
 Detected units: M2

Raw source area fields were not modified.
 Derived publication km² fields were created explicitly.

15.3 Final tree–settlement class distributions

Retained QC output

STEP 2 – FINAL TREE-SETTLEMENT DISTRIBUTIONS

DIOCESES

Tree-Dominated Landscape	719
Tree-Rich Settled Landscape	513
Mixed Tree-Settlement Landscape	721
Built Landscape with Substantial Tree Cover	157
Built Landscape with Limited Tree Cover	166
Open or Sparsely Treed Landscape	598

EPISCOPAL CONFERENCES

Tree-Dominated Landscape	41
Tree-Rich Settled Landscape	15
Mixed Tree-Settlement Landscape	44
Built Landscape with Substantial Tree Cover	3
Built Landscape with Limited Tree Cover	2
Open or Sparsely Treed Landscape	17

LSAP MEMBERS – 20 KM	
Tree-Dominated Landscape	81
Tree-Rich Settled Landscape	357
Mixed Tree-Settlement Landscape	214
Built Landscape with Substantial Tree Cover	642
Built Landscape with Limited Tree Cover	1,260
Open or Sparsely Treed Landscape	185

LSAP MEMBERS – 100 KM	
Tree-Dominated Landscape	236
Tree-Rich Settled Landscape	592
Mixed Tree-Settlement Landscape	619
Built Landscape with Substantial Tree Cover	613
Built Landscape with Limited Tree Cover	257
Open or Sparsely Treed Landscape	422

Invalid percentage values: 0 at every scale.

Invalid km² areas: 0 at every scale.

15.4 Member transition QC

Retained QC output

STEP 3 – LSAP MEMBER 20 KM -> 100 KM TRANSITION

Pairing:

20 km BUFFER_ID values: 2,739

100 km BUFFER_ID values: 2,739

Only in 20 km: 0

Only in 100 km: 0

PASS: exact 1:1 pairing.

TREE CHANGE (regional minus local; classified cover)

10th -4.06 | 25th 1.24 | median 8.73 | 75th 21.03 | 90th 27.82

min -39.76 | max 55.45

BUILT CHANGE

10th -48.36 | 25th -35.78 | median -22.91 | 75th -7.65 | 90th -0.80

min -64.93 | max 12.89

Comparison coverage:

High-Confidence Comparison	1,881
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Moderate-Confidence Comparison	352
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Limited-Coverage Comparison	459
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Very Limited-Coverage Comparison	47
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Most common context shifts:

458 Built Limited -> Built with Substantial Tree Cover

291 Built with Substantial Tree Cover -> Tree-Rich Settled

286 Built Limited -> Open/Sparsely Treed

248 Built Limited at both scales

205 Built Limited -> Mixed Tree-Settlement

198 Tree-Rich Settled at both scales

15.5 Conference LSAP activity audit

Retained QC output

STEP 3B – CONFERENCE LSAP ACTIVITY AUDIT

Enrolled members:

nonnull 115 | positive 109 | total 12,815 | median 17 | max 2,879

Plans:

positive 65 | total 1,418 | median 1 | max 444

Actions:

LSAP Integral Ecology Atlas | Trees, Cities and the Landscapes Around Us | Technical Report

positive 53 | total 8,556 | median 0 | max 1,701

Reflections:

positive 60 | total 1,181 | median 1 | max 295

Related Laudato Si' Goals:

Cry of the Earth:	total 1,115
Sustainable Lifestyles:	total 1,794
Ecological Education:	total 1,243
Community Resilience:	total 1,161
Ecological Spirituality:	total 1,217

Participating sectors:

Parishes enrolled:	863
Dioceses enrolled:	322
Schools enrolled:	1,366
Universities enrolled:	303
Religious communities:	702
Religious congregations:	825
Organizations & groups:	1,193
Health centers:	165
Economic entities:	117

15.6 Story completeness

Retained QC output

STEP 4 – STORY COMPLETENESS

DIOCESES

8 initial story fields populated for all 2,874 records.

EPISCOPAL CONFERENCES

9 initial story fields populated for all 122 records.

LSAP MEMBERS – 20 KM

9 initial story fields populated for all 2,739 records.

LSAP MEMBERS – 100 KM

9 initial story fields populated for all 2,739 records.

STEP 4B

STORY_INTEGRATED_SUMMARY missing = 0 at every scale

STORY_SOLUTIONS missing = 0 at every scale

15.7 Canopy-height and BII class distributions

Retained QC output

STEP 4D – SCIENCE INTERPRETATION CLASSES

DIOCESES – CANOPY HEIGHT

Short 558 | Lower 737 | Moderate 676 | Tall 525 | Very Tall 228 | >=25 m 150

DIOCESES – BII

High 1,130 | Relatively High 1,222 | Intermediate 370 | Reduced 108 | Substantially Reduced 44

EPISCOPAL CONFERENCES – CANOPY HEIGHT

Short 23 | Lower 32 | Moderate 26 | Tall 25 | Very Tall 11 | >=25 m 5

EPISCOPAL CONFERENCES – BII

High 49 | Relatively High 60 | Intermediate 5 | Reduced 6 | Substantially Reduced 2

LSAP 20 KM – CANOPY HEIGHT

Short 361 | Lower 732 | Moderate 754 | Tall 629 | Very Tall 224 | >=25 m 39

LSAP 20 KM – BII

High 524 | Relatively High 1,114 | Intermediate 910 | Reduced 115 | Substantially Reduced 76

LSAP 100 KM – CANOPY HEIGHT

Short 336 | Lower 681 | Moderate 798 | Tall 593 | Very Tall 286 | >=25 m 45

LSAP 100 KM – BII

High 695 | Relatively High 1,439 | Intermediate 508 | Reduced 82 | Substantially Reduced 15

No missing canopy classes, BII classes, or STORY_ECOLOGY values.

15.8 Final prose QC

Retained QC output

STEP 4H – FINAL PROSE POLISH

DIOCESES

'a open...' remaining: 0
'1 parishes' remaining: 0
religious-order double periods: 0
STORY_INTEGRATED_SUMMARY missing: 0
STORY_SOLUTIONS missing: 0

EPISCOPAL CONFERENCES

same checks: all 0

LSAP MEMBERS – 20 KM

same checks: all 0

LSAP MEMBERS – 100 KM

same checks: all 0

Full technical caveats were removed from public-facing integrated narratives and retained in STORY_DATA_NOTE.

15.9 Final publication cleanup

Retained QC output

STEP 5 – FINAL PUBLICATION CLEANUP

DIOCESES

User fields before cleanup: 160
Removed: 101
Final user fields: 59
Required fields missing: 0
Story fields: 10 | missing values: 0

EPISCOPAL CONFERENCES

Before: 304
Removed: 230
Final: 74
Required fields missing: 0
Story fields: 11 | missing values: 0

LSAP MEMBERS – 20 KM

Before: 146
Removed: 85
Final: 61
Required fields missing: 0
Story fields: 11 | missing values: 0

LSAP MEMBERS – 100 KM

Before: 146
Removed: 85
Final: 61
Required fields missing: 0
Story fields: 11 | missing values: 0

Final group:

Trees, Cities and the Landscapes Around Us

Final layer order:

LSAP Integral Ecology Atlas | Trees, Cities and the Landscapes Around Us | Technical Report

16. Final publication schemas and field counts

Publication cleanup removes the large biodiversity-summary schema while retaining only fields needed for identity, science, interpretation, Church/LSAP context, cross-scale comparison, and storytelling.

Layer	Final user fields	Story fields	Required-field QC
Dioceses	59	10	PASS
Episcopal Conferences	74	11	PASS
LSAP Members — 20 km	61	11	PASS
LSAP Members — 100 km	61	11	PASS

16.1 Core story fields

- STORY_SUMMARY — concise human-centered popup summary.
- STORY_INTEGRATED_SUMMARY — longer public narrative bringing identity, landscape, science, scale relationship, and stewardship together.
- STORY_ECOLOGY — detailed modeled canopy-height and BII interpretation.
- STORY_SOLUTIONS — landscape-class-specific stewardship possibilities.
- STORY_DATA_NOTE — technical source and uncertainty caveats.
- STORY_CHURCH — diocesan/Conference Catholic scale and leadership.
- STORY_LSAP — Conference-level LSAP participation, activity, goals, and sectors.
- STORY_EXPLORE_LOCAL — Conference guidance toward diocesan/member scales.
- STORY_LOCAL_TO_REGION — member 20 km → 100 km transition narrative.

17. Scientific limitations and interpretation rules

- Tree land-cover share and modeled canopy height measure different things and must not be conflated.
- Canopy height is modeled, spatially summarized, and subject to source uncertainties; ≥25 m receives additional caution.
- BII is a modeled indicator relative to near-undisturbed reference conditions, not a direct field census and not a simple 'health score.'
- The 0.90 BII threshold has external significance; other BII text classes are project descriptive categories.
- Tree-settlement categories and canopy-height classes are project-derived interpretive tools, not source-provider classes.
- Low tree cover is not necessarily an ecological deficit; open ecosystems may be the correct native condition.
- WorldPop and Catholic population counts are independent estimates and are not interchangeable.
- 100 km member analysis geographies have variable represented area and sometimes incomplete land-cover coverage.
- Cross-scale changes use normalized classified cover to reduce coverage bias.
- Episcopal Conference solutions are intentionally broad; local action should be interpreted through dioceses and member ecological neighborhoods.
- Ecclesiastical boundaries are organizational geographies, not ecological boundaries.

18. ArcGIS Online-ready metadata text

18.1 Map summary

Trees, Cities and the Landscapes Around Us explores how Catholic communities and Laudato Si' Action Platform participants are situated within tree–settlement landscapes. The analysis combines tree and built land-cover shares, modeled tree canopy height, WorldPop population context, biodiversity intactness, Catholic statistics, and LSAP participation across dioceses, Episcopal Conferences, and 20 km / 100 km member landscapes.

18.2 Description

This map is part of the LSAP Integral Ecology Atlas. Environmental indicators are derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons and summarized to Catholic ecclesiastical geographies and LSAP member-centered analysis areas. Tree–settlement classes use tree and built shares normalized to classified land cover; raw percentages remain available for public interpretation. Modeled canopy height is treated as a structural tree indicator, and Biodiversity Intactness Index values are interpreted relative to near-undisturbed ecological reference conditions. The map connects environmental science with Catholic population/parish context, Church leadership, LSAP participation, local-to-regional landscape transitions, and scale-appropriate stewardship possibilities.

18.3 Suggested tags

Trees, Tree Canopy Height, Cities, Built Environment, Land Cover, Biodiversity, Biodiversity Intactness, WorldPop, H3, Esri Living Atlas, Laudato Si', Laudato Si' Action Platform, LSAP, Integral Ecology, Catholic Church, Dioceses, Episcopal Conferences, Parishes, Church Geography, Urban Ecology, Green Infrastructure, Habitat Connectivity, Stewardship

18.4 Compact attribution

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas Catholic jurisdictional framework; Holy See statistical publications; Catholic-Hierarchy.org. LSAP member/activity data: Laudato Si' Action Platform / Dicastery for Promoting Integral Human Development. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and underlying credited sources, including Esri Sentinel-2 Land Use/Land Cover, WRI/Meta tree canopy height, Natural History Museum/PREDICTS Biodiversity Intactness Index, and WorldPop.

19. References

- Esri. (2026, July 8). Global Biodiversity and Conservation Hexagons... but better! ArcGIS Blog. <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>
- Pisut, D., & VanGraafeiland, K. (2025, July 9). Introducing Global Biodiversity and Conservation Hexagons. ArcGIS Blog. <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/biodiversity-conservation-hexagons>
- World Resources Institute, Land & Carbon Lab. High Resolution Canopy Height Map. WRI Data Explorer. Last updated October 6, 2025. <https://datasets.wri.org/datasets/meta-tree-canopy-height>
- Natural History Museum. Biodiversity Intactness Index. PREDICTS biodiversity indicators. <https://www.nhm.ac.uk/our-science/services/data/biodiversity-intactness-index.html>
- WorldPop. Gridded population estimate datasets and tools. <https://www.worldpop.org/methods/populations/>
- WorldPop. Global2: Global high-resolution population estimates for 2015–2030. <https://www.worldpop.org/blog/worldpop-global2-global-high-resolution-population-estimates-for-2015-2030/>

Project sources: LSAP_Diocese_BiodiversitySummary; LSAP_EpiscopalConferences_BiodiversitySummary; LSAP_MemberPoints_20km_BiodiversitySummary; LSAP_MemberPoints_100km_BiodiversitySummary; project Catholic jurisdictional framework; Laudato Si' Action Platform member/activity extract; Holy See statistical publications; Catholic-Hierarchy.org.

Appendix A. Reproducible ArcPy science code

The code below consolidates the final analytical logic used in the completed workflow. It is intended as a reproducibility reference. The executed ArcGIS Pro notebook also contains intermediate audits and story-generation utilities.

A.1 Core classification and area helpers

```
import arcpy
import statistics

VALID_LULC = [
    "PCT_LULC_Water",
    "PCT_LULC_Trees",
    "PCT_LULC_Crops",
    "PCT_LULC_Built",
    "PCT_LULC_Rangeland",
    "PCT_LULC_Bare",
    "PCT_LULC_SnowIce",
    "PCT_LULC_Veg"
]

def safe(v):
    return 0.0 if v is None else float(v)

def diagnose_area_units(fc, field):
    values = []
    with arcpy.da.SearchCursor(fc, [field]) as cursor:
        for (value,) in cursor:
            if value is not None and float(value) > 0:
                values.append(float(value))

    median_value = statistics.median(values)

    if median_value >= 1_000_000:
        return "M2", median_value

    return "KM2", median_value

def to_km2(value, units):
    if value is None:
        return None

    value = float(value)

    if units == "M2":
        return value / 1_000_000.0

    return value

def tree_context(tree_class_pct):
    if tree_class_pct >= 40:
        return "Tree-Rich"
    if tree_class_pct >= 15:
        return "Mixed Tree Cover"
    return "Lower Tree Cover"

def settlement_context(built_class_pct):
    if built_class_pct < 5:
        return "Lightly Built"
    if built_class_pct < 15:
        return "Settled"
    return "Highly Built"

def tree_settlement_context(tree_class_pct, built_class_pct):
    tree_group = tree_context(tree_class_pct)
    built_group = settlement_context(built_class_pct)

    if tree_group == "Tree-Rich":
```

```

        if built_group == "Lightly Built":
            return "Tree-Dominated Landscape"
        return "Tree-Rich Settled Landscape"

    if tree_group == "Mixed Tree Cover":
        if built_group == "Highly Built":
            return "Built Landscape with Substantial Tree Cover"
        return "Mixed Tree-Settlement Landscape"

    if built_group == "Highly Built":
        return "Built Landscape with Limited Tree Cover"

    return "Open or Sparsely Treed Landscape"

def coverage_context(coverage):
    if coverage >= 95:
        return "Near-Complete Coverage"
    if coverage >= 90:
        return "High Coverage"
    if coverage >= 75:
        return "Moderate Coverage"
    if coverage >= 50:
        return "Limited Coverage"
    return "Very Limited Coverage"

```

A.2 Foundation calculation pattern

```

# The raw source area field remains untouched.
units, median_value = diagnose_area_units(fc, source_area_field)

with arcpy.da.UpdateCursor(fc, cursor_fields) as cursor:
    for row in cursor:

        landscape_km2 = to_km2(
            row[IDX[source_area_field]],
            units
        )

        tree_pct = safe(row[IDX["PCT_LULC_Trees"]])
        built_pct = safe(row[IDX["PCT_LULC_Built"]])
        veg_pct = safe(row[IDX["PCT_LULC_Veg"]])

        tree_area = safe(
            to_km2(
                row[IDX["TOTAL_LULC_Trees"]],
                units
            )
        )

        built_area = safe(
            to_km2(
                row[IDX["TOTAL_LULC_Built"]],
                units
            )
        )

        veg_area = safe(
            to_km2(
                row[IDX["TOTAL_LULC_Veg"]],
                units
            )
        )

        valid_cover = sum(
            max(0.0, safe(row[IDX[field]]))
            for field in VALID_LULC
        )

        cloud_nodata = (
            max(0.0, safe(row[IDX["PCT_LULC_Cloud"]])) +
            max(0.0, safe(row[IDX["PCT_LULC_NoData"]]))

```

```

)

other_unmapped = max(
    0.0,
    100.0 - (valid_cover + cloud_nodata)
)

unclassified = max(
    0.0,
    100.0 - valid_cover
)

if valid_cover > 0:
    tree_class_pct = tree_pct / valid_cover * 100.0
    built_class_pct = built_pct / valid_cover * 100.0
else:
    tree_class_pct = 0.0
    built_class_pct = 0.0

context = tree_settlement_context(
    tree_class_pct,
    built_class_pct
)

pop_2025 = safe(
    row[IDX["TOTAL_WorldPop_2025"]]
)

pop_density = (
    pop_2025 / landscape_km2
    if landscape_km2 and landscape_km2 > 0
    else None
)

# Write explicit publication derivatives:
# LANDSCAPE_KM2
# TREE_LAND_PCT
# BUILT_PCT
# OTHER_VEG_PCT
# TREE_VEG_PCT
# TREE_AREA_KM2
# BUILT_AREA_KM2
# VEG_AREA_KM2
# TREE_VEG_AREA_KM2
# VALID_COVER_PCT
# UNCLASSIFIED_PCT
# CLOUD_NODATA_PCT
# OTHER_UNMAPPED_PCT
# TREE_CLASS_PCT
# BUILT_CLASS_PCT
# TREE_SETTLEMENT_CONTEXT
# CANOPY_MEAN
# POP_2025
# POP_DENSITY
# BIODIV_INTACT

```

A.3 Member 20 km → 100 km transition

```

def comparison_coverage(local_cov, regional_cov):
    minimum = min(float(local_cov), float(regional_cov))

    if minimum >= 90:
        return "High-Confidence Comparison"
    if minimum >= 75:
        return "Moderate-Confidence Comparison"
    if minimum >= 50:
        return "Limited-Coverage Comparison"
    return "Very Limited-Coverage Comparison"

# BUFFER_ID is the authoritative pairing key.

```

```

tree_change = (
    regional_record["TREE_CLASS_PCT"] -
    local_record["TREE_CLASS_PCT"]
)

built_change = (
    regional_record["BUILT_CLASS_PCT"] -
    local_record["BUILT_CLASS_PCT"]
)

if local_record["TREE_SETTLEMENT_CONTEXT"] == regional_record["TREE_SETTLEMENT_CONTEXT"]:
    shift = (
        f"{local_record['TREE_SETTLEMENT_CONTEXT']} at both scales"
    )
else:
    shift = (
        f"{local_record['TREE_SETTLEMENT_CONTEXT']} → "
        f"{regional_record['TREE_SETTLEMENT_CONTEXT']}"
    )

comparison_class = comparison_coverage(
    local_record["VALID_COVER_PCT"],
    regional_record["VALID_COVER_PCT"]
)

```

A.4 Modeled canopy-height interpretation

```

def canopy_height_class(v):
    if v is None:
        return "No Canopy-Height Estimate"
    if v < 5:
        return "Short Canopy Structure"
    if v < 10:
        return "Lower Canopy Structure"
    if v < 15:
        return "Moderate Canopy Structure"
    if v < 20:
        return "Tall Canopy Structure"
    if v < 25:
        return "Very Tall Canopy Structure"
    return "Exceptionally Tall Modeled Canopy"

def canopy_height_story(v):
    category = canopy_height_class(v)

    if v >= 25:
        caution = (
            " Values at and above 25 m should be interpreted with "
            "additional caution because the source model was not "
            "field-validated for very tall trees above 25 m."
        )
    else:
        caution = ""

    return (
        f"Modeled mean tree canopy height is approximately {v:.1f} m, "
        f"classified here as {category.lower()}. "
        f"This is a spatially summarized modeled indicator, not the "
        f"measured height of every tree."
        f"{caution}"
    )

```

A.5 Biodiversity Intactness interpretation

```

def intactness_class(v):
    if v is None:
        return "No Biodiversity-Intactness Estimate"
    if v >= 0.90:
        return "High Modeled Biodiversity Intactness"
    if v >= 0.80:
        return "Relatively High Modeled Biodiversity Intactness"

```

```

if v >= 0.70:
    return "Intermediate Modeled Biodiversity Intactness"
if v >= 0.60:
    return "Reduced Modeled Biodiversity Intactness"
return "Substantially Reduced Modeled Biodiversity Intactness"

def intactness_story(v):
    category = intactness_class(v)

    if v >= 1.0:
        meaning = (
            "very high / near-reference modeled intactness rather than "
            "more than 100% intact"
        )
    elif v >= 0.90:
        meaning = (
            "high modeled intactness; values above 0.90 fall within "
            "the proposed planetary-boundary range"
        )
    elif v >= 0.80:
        meaning = (
            "relatively high modeled intactness while still reflecting "
            "ecological change from near-undisturbed conditions"
        )
    elif v >= 0.70:
        meaning = (
            "an intermediate level of modeled intactness relative to "
            "near-undisturbed reference conditions"
        )
    else:
        meaning = (
            "substantial modeled departure from near-undisturbed "
            "reference conditions"
        )

    return (
        f"The modeled mean Biodiversity Intactness Index is {v:.2f}, "
        f"classified here as {category.lower()}, indicating {meaning}. "
        f"The index should not be read as a simple '{v*100:.0f}% healthy' "
        f"score or as a literal count of species remaining."
    )

```

A.6 Conference LSAP activity selection

```

RELATED_GOALS = [
    ("Response to the Cry of the Earth",
     "USER_Response_to_the_Cry_of_the_Earth"),

    ("Adoption of Sustainable Lifestyles",
     "USER_Adoption_of_sustainable_lifestyles"),

    ("Ecological Education",
     "USER_Ecological_education"),

    ("Community Resilience & Empowerment",
     "USER_Community_resilience_and_empowerment"),

    ("Ecological Spirituality",
     "USER_Ecological_spirituality")
]

RELATED_SECTORS = [
    ("parishes", "USER_Parish_Enrolled_Cnt"),
    ("dioceses", "USER_Diocese_Enrolled_Cnt"),
    ("schools", "USER_School_Enrolled_Cnt"),
    ("universities", "USER_Uni_Enrolled_Cnt"),
    ("religious communities", "USER_Religious_Com_Enrolled_Cnt"),
    ("religious congregations", "USER_ReligiousCong_Enrolled_Cnt"),
    ("organizations and groups", "USER_OrgsGroups_Enrolled_Cnt"),
    ("health centers", "USER_HealthInst_Enrolled_Cnt"),
    ("economic entities", "USER_EconEn_Enrolled_Cnt")
]

```

```
]

# For each Conference:
# 1. Anchor with USER_Enrolled_total.
# 2. Add plans/actions/reflections when positive.
# 3. Rank positive related goals and report top three.
# 4. Rank positive participating sectors and report top three.
# 5. If no usable summary exists, say so explicitly rather than inventing one.
```

A.7 Human-centered story ordering

```
# MEMBER integrated story order:
integrated = " ".join(
    text
    for text in [
        identity,      # Who is here?
        landscape,     # What kind of place?
        science,       # What do the indicators mean?
        transition,    # How does 20 km differ from 100 km?
        solutions,     # What stewardship directions fit this context?
    ]
    if text
)

# Technical source/model caveats remain in STORY_DATA_NOTE rather than
# overwhelming the public-facing integrated narrative.
```

A.8 Publication cleanup pattern

```
# Publication derivatives keep only identity, final science,
# Church/LSAP context, transition metrics, and story fields.

existing = {f.name for f in arcpy.ListFields(fc)}

publication_keep = KEEP_FIELDS & existing

delete_fields = [
    f.name
    for f in arcpy.ListFields(fc)
    if (
        not f.required and
        f.name not in publication_keep
    )
]

if delete_fields:
    arcpy.management.DeleteField(
        fc,
        delete_fields
    )

# Always follow with:
# - required-field check
# - story completeness check
# - final user-field count
# - map-placement verification
```

Appendix B. Retained QC logs

The full executed outputs were preserved during development. This appendix repeats the final high-value checks that support publication status.

Retained QC output

FINAL PUBLICATION STATUS

DIOCESES

2,874 features
59 final user fields
Required fields missing: 0
Story fields: 10
Missing story values: 0

EPISCOPAL CONFERENCES

122 features
74 final user fields
Required fields missing: 0
Story fields: 11
Missing story values: 0

LSAP MEMBERS – 20 KM

2,739 features
61 final user fields
Required fields missing: 0
Story fields: 11
Missing story values: 0

LSAP MEMBERS – 100 KM

2,739 features
61 final user fields
Required fields missing: 0
Story fields: 11
Missing story values: 0

ANALYTICAL STATUS

Raw source area fields unchanged.
Explicit km² publication derivatives created.
Tree-settlement classes use normalized classified cover.
Canopy height interpreted as modeled height, not canopy-cover percentage.
BII interpreted relative to near-undisturbed reference conditions.
Conference LSAP activity included.
Conference local-action prescriptions avoided.
Member 20 km / 100 km transitions paired exactly by BUFFER_ID.
Final public integrated stories begin with Church/member identity.
Technical caveats retained separately in STORY_DATA_NOTE.

PUBLICATION GROUP

Trees, Cities and the Landscapes Around Us

ORDER

LSAP Members – 20 km Local Ecological Neighborhood
LSAP Members – 100 km Regional Landscape Context
Dioceses – Tree-Settlement Landscapes
Episcopal Conferences – Tree-Settlement Landscapes